

TASK 1

```
import random

# Part A – Machine Initialization
machines = ['A', 'B', 'C']

# Initial values
N = {'A': 0, 'B': 0, 'C': 0}      # Number of selections
Q = {'A': 0, 'B': 0, 'C': 0}      # Q-values
R = {'A': 0, 'B': 0, 'C': 0}      # Total rewards

epsilon = 0.2
N_play = 10

# Play for 10 times
for play in range(1, N_play + 1):

    # Generate random number
    r = random.random()

    # Part C – E-Greedy Decision
    if r < epsilon:
        # Exploration
        machine = random.choice(machines)
        decision = "Exploration"
    else:
        # Exploitation
        machine = max(Q, key=Q.get)
        decision = "Exploitation"

    # Part D – Reward Collection
    reward = float(input(f"Play {play}: Play with machine {machine}\n"))

    # Part E – Q-Value Update
    N[machine] += 1

    Q[machine] = Q[machine] + (
        reward - Q[machine]
    ) / N[machine]

    R[machine] += reward

    print(f"{decision} -> Machine {machine}")
    print(f"Updated Q({machine}) = {Q[machine]:.2f}\n")
```

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# Part F – Display Results
print("\n===== FINAL RESULTS =====")

for machine in machines:
    print(f"Machine {machine}")
    print(f"N({machine}) = {N[machine]}")
    print(f"Total Reward = {R[machine]}")
    print(f"Q({machine}) = {Q[machine]:.2f}")
    print()

print("Final Q-values:", Q)
```

Play 3: Play with machine A
Enter reward: 3
Exploitation -> Machine A
Updated Q(A) = 2.00

Play 4: Play with machine A
Enter reward: 4
Exploitation -> Machine A
Updated Q(A) = 2.50

Play 5: Play with machine A
Enter reward: 6
Exploitation -> Machine A
Updated Q(A) = 3.20

Play 6: Play with machine A
Enter reward: 4
Exploration -> Machine A
Updated Q(A) = 3.33

Play 7: Play with machine A
Enter reward: 4
Exploitation -> Machine A
Updated Q(A) = 3.43

Play 8: Play with machine A
Enter reward: 4
Exploitation -> Machine A
Updated Q(A) = 3.50

Play 9: Play with machine A
Enter reward: 4
Exploitation -> Machine A
Updated Q(A) = 3.56

Play 10: Play with machine C
Enter reward: 23

Exploration -> Machine C
Updated $Q(C) = 23.00$

===== FINAL RESULTS =====

Machine A

$N(A) = 9$

Total Reward = 32.0

$Q(A) = 3.56$

Machine B

$N(B) = 0$

Total Reward = 0

$Q(B) = 0.00$

Machine C

$N(C) = 1$

Total Reward = 23.0

$Q(C) = 23.00$

Final Q values: {'A': 3.5555555555555554, 'B': 0, 'C': 23.0}

TASK 2

```
import random

# Machines
machines = ['A', 'B', 'C']

# Reward patterns
rewards = {
    'A': [3, 4, 2, 5],
    'B': [6, 7, 5, 8],
    'C': [4, 3, 5, 4]
}

# Tables
N = {'A': 0, 'B': 0, 'C': 0}
Q = {'A': 0, 'B': 0, 'C': 0}

epsilon = 0.2
reward_index = {'A': 0, 'B': 0, 'C': 0}

# 10 plays
for play in range(1, 11):

    r = random.random()

    # Epsilon-greedy selection
    if r < epsilon:
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        machine = random.choice(machines)
        decision = "Exploration"
    else:
        machine = max(Q, key=Q.get)
        decision = "Exploitation"

    # Get reward from reward pattern
    index = reward_index[machine]

    # If all rewards are used, start again
    if index >= len(rewards[machine]):
        index = 0

    reward = rewards[machine][index]

    reward_index[machine] += 1

    # Update N
    N[machine] += 1

    # Incremental Q-value update
    Q[machine] = Q[machine] + (
        reward - Q[machine]
    ) / N[machine]

    print(
        f"Play {play}: "
        f"{decision}, "
        f"Machine {machine}, "
        f"Reward {reward}, "
        f"Q = {Q[machine]:.2f}"
    )

# Final results
print("\n===== FINAL RESULTS =====")

total_reward = 0

for machine in machines:
    print(f"\nMachine {machine}")
    print(f"N({machine}) = {N[machine]}")
    print(f"Q({machine}) = {Q[machine]:.2f}")

print("\nFinal Q Table:")
print(Q)

print("\nTotal reward received after 10 plays:")

```

```
print("Calculate by adding all rewards printed above.")
```

```
Play 1: Exploitation, Machine A, Reward 3, Q = 3.00
Play 2: Exploitation, Machine A, Reward 4, Q = 3.50
Play 3: Exploitation, Machine A, Reward 2, Q = 3.00
Play 4: Exploration, Machine A, Reward 5, Q = 3.50
Play 5: Exploration, Machine B, Reward 6, Q = 6.00
Play 6: Exploitation, Machine B, Reward 7, Q = 6.50
Play 7: Exploitation, Machine B, Reward 5, Q = 6.00
Play 8: Exploitation, Machine B, Reward 8, Q = 6.50
Play 9: Exploitation, Machine B, Reward 6, Q = 6.40
Play 10: Exploitation, Machine B, Reward 6, Q = 6.33
```

```
===== FINAL RESULTS =====
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```
Machine A
N(A) = 4
Q(A) = 3.50
```

```
Machine B
N(B) = 6
Q(B) = 6.33
```

```
Machine C
N(C) = 0
Q(C) = 0.00
```

```
Final Q Table:
{'A': 3.5, 'B': 6.333333333333334, 'C': 0}
```

```
Total reward received after 10 plays:
Calculate by adding all rewards printed above.
```

